

picoCTF Forensics - Sleuthkit Apprentice (disk.flag.img)

Challenge: Find the flag hidden in a compressed disk image using sleuthkit tools.
Tools Used: file, binwalk, fls, icat, mmls, strings

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Challenge Overview

Field	Value
Challenge Name	Sleuthkit Apprentice (disk.flag.img variant)
Category	Forensics
File	disk.flag.img.gz
Goal	Find flag.txt or flag.uni.txt inside the disk image
Difficulty	Medium

Step-by-Step Solution

Step 1: Decompress the Image

```
gunzip disk.flag.img.gz
```

Result:

```
disk.flag.img      (uncompressed raw disk image)
```

Sleuthkit tools require raw (uncompressed) disk images to work properly.

Step 2: Identify File Type

```
file disk.flag.img
```

Output:

```
disk.flag.img: DOS/MBR boot sector; partition 1 : ID=0x83, active, start-CHS (0x0,32,33), end-CHS
↪  (0x10,81,1), startsector 2048, 260096 sectors
```

Key Insight: The image contains a **partition table** (DOS/MBR), not just a raw filesystem. This means we need to find the correct partition offset before we can access files.

Step 3: Analyze Partitions with binwalk

```
binwalk disk.flag.img
```

Output:

DECIMAL	HEXADECIMAL	DESCRIPTION

1048576	0x100000	Linux EXT filesystem, blocks count: 102400, image size: 104857600, rev
↪ 1.0,		ext4 filesystem data, UUID=0fcf6c07-4376-4be0-b7da-e7b455395539
184549376	0xB000000	Linux EXT filesystem, blocks count: 126976, image size: 130023424, rev
↪ 1.0,		ext4 filesystem data, UUID=55c2d6cb-32d6-4b22-b398-28f036413641

Discovery: There are **TWO** ext4 partitions:

Partition	Decimal Offset	Hex Offset	Description
1st	1,048,576	0x100000	Boot partition (~100 MB)
2nd	184,549,376	0xB000000	Main filesystem (~130 MB)

Step 4: Calculate Offsets

Sleuthkit fls and icat require offsets in **sectors**, not bytes.

Formula:

```
Sector Offset = Byte Offset / 512
```

Calculations:

Partition	Byte Offset	Sector Offset
1st	1,048,576	1,048,576 / 512 = 2048
2nd	184,549,376	184,549,376 / 512 = 360,448

Step 5: List Files in Each Partition

First Partition (Boot)

```
fls -o 2048 disk.flag.img
```

Output:

```
d/d 11: lost+found
r/r 12: ldlinux.sys
r/r 13: ldlinux.c32
r/r 15: config-virt
r/r 16: vmlinuz-virt
r/r 17: initramfs-virt
l/l 18: boot
r/r 20: libutil.c32
r/r 19: extlinux.conf
r/r 21: libcom32.c32
r/r 22: mboot.c32
r/r 23: menu.c32
r/r 14: System.map-virt
r/r 24: vesamenu.c32
V/V 25585:      $OrphanFiles
```

These are all bootloader files. No flag here.

Second Partition (Main Filesystem)

```
fls -o 360448 disk.flag.img
```

Output:

```
d/d 451:      home
d/d 11: lost+found
d/d 12: boot
d/d 1985:     etc
d/d 1986:     proc
d/d 1987:     dev
d/d 1988:     tmp
d/d 1989:     lib
d/d 1990:     var
d/d 3969:     usr
d/d 3970:     bin
d/d 1991:     sbin
d/d 1992:     media
d/d 1993:     mnt
d/d 1994:     opt
d/d 1995:     root
d/d 1996:     run
d/d 1997:     srv
d/d 1998:     sys
d/d 2358:     swap
```

```
V/V 31745:      $OrphanFiles
```

This looks like a full Linux root filesystem. The flag is likely in /root or /home.

Step 6: Navigate to the Flag

Check root's home directory

```
fls -o 360448 disk.flag.img 1995
```

Output:

```
r/r 2363:      .ash_history
d/d 3981:      my_folder
```

Check my_folder

```
fls -o 360448 disk.flag.img 3981
```

Output:

```
r/r * 2082(realloc):  flag.txt
d/d 2371:      flag.uni.txt
```

Found! There are two flag files:

- flag.txt (deleted/reallocated inode 2082)
- flag.uni.txt (inode 2371)

Step 7: Extract the Flag

```
# Extract flag.uni.txt (active file)
icat -o 360448 disk.flag.img 2371
```

Output:

```
picoCTF{f0r3nslc4t0r_n0v1c3_3nd0r_4bd721f2}
```

Alternatively, try the deleted file:

```
icat -o 360448 disk.flag.img 2082
```

Sleuthkit Commands Reference

Command	Purpose	Example
<code>fls -o OFFSET image</code>	List files in root directory	<code>fls -o 360448 disk.flag.img</code>
<code>fls -o OFFSET image INODE</code>	List files in a directory	<code>fls -o 360448 disk.flag.img 1995</code>
<code>icat -o OFFSET image INODE</code>	Extract file by inode	<code>icat -o 360448 disk.flag.img 2371</code>

Table 4 – continued

Command	Purpose	Example
<code>fsstat -o OFFSET image</code>	Show filesystem info	<code>fsstat -o 360448 disk.flag.img</code>
<code>istat -o OFFSET image INODE</code>	Show inode details	<code>istat -o 360448 disk.flag.img</code> 2371
<code>mmls image</code>	Show partition table	<code>mmls disk.flag.img</code>
<code>srch_strings image</code>	Search for strings	<code>srch_strings disk.flag.img</code>

Common Errors & Fixes

Error	Cause	Solution
Cannot determine file system type	Missing offset or wrong offset	Use <code>-o</code> flag with correct sector offset
File not found	Wrong inode number	Use <code>fls</code> to find correct inode
Empty output from <code>icat</code>	File is deleted or wrong inode	Try <code>istat</code> to check inode status
No partitions found	Image not decompressed	Run <code>gunzip</code> first

Key Takeaways

1. **Always decompress first** - Sleuthkit needs raw images
2. **Check for multiple partitions** - Use `binwalk` or `mmls`
3. **Calculate sector offsets** - Byte Offset / 512
4. **Use `-o` flag** - Required for partitioned images
5. **Search all partitions** - The flag may not be in the first one
6. **Check deleted files** - `r/r * inode(realloc)` may still contain data

Flag

```
picoCTF{f0r3nslc4t0r_n0v1c3_3nd0r_4bd721f2}
```

Tools Installation

```
# Debian/Ubuntu
sudo apt-get install sleuthkit binwalk

# Arch
sudo pacman -S sleuthkit binwalk

# macOS
brew install sleuthkit binwalk
```

Happy Forensics!